

Elastic Bands as a Component of Periodized Resistance Training

Running Head: Elastic Bands with Periodized Training

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Abstract

Variable resistance training has recently become a component of strength and conditioning programs. Prior research has demonstrated increases in power and/or strength using low loads of variable resistance. However, no study has examined using high loads of variable resistance as a part of a periodized training protocol. **Purpose:** to examine variable resistance training within the context of a periodized training program, and to examine a greater load of variable resistance than has been examined in prior research. **Methods:** 14 NCAA Division II male basketball players were recruited for this study. Athletes were divided equally into either a variable resistance or control group. The variable resistance group added 30% of their one repetition maximum as band tension to their prescribed weight one session per week. Rate of power development, peak power, strength, body composition, and vertical jump height were measured pre and post treatment. **Results:** No baseline differences were observed between groups for any measurement of strength, power, or body composition. A significant group by time interaction was observed for RPD, in which RPD was greater in VRT post training than in the control group. Significant time effects were observed for all other variables including squat 1RM, bench press 1RM, deadlift 1-RM, clean 3-RM, vertical jump, and lean mass. While there were no significant group X time interactions, the VRT group's percent changes and ESs indicate a larger treatment effect in the squat and bench press 1RM values and the vertical jump performed on the force plate and vertec. **Conclusions:** These results suggest that when using variable resistance as a component of a periodized training program, power and strength can

1 be enhanced. Therefore, athletes whom add variable resistance to one training session per
2 week may enhance their athletic performance.

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4 KEY WORDS: bands; periodization; power; strength; variable resistance

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Introduction

Strength and power are two of the most critical attributes underlying success in sport (15, 21).

Moreover, the rate that an individual develops power (RPD) has been recognized as being equal, or perhaps of greater importance, than power itself (14). Subsequently, these attributes are focal points of many strength and conditioning programs. A variety of training techniques are therefore employed to augment strength, power, and RPD adaptations (5, 6). One such technique is the use of elastic bands, or variable resistance training (VRT), as a part of a resistance-training program (1, 2, 7, 8, 11, 12, 14, 19, 20).

Variable resistance training has been used in the sport of competitive powerlifting for over a decade (18), and more recently, they have become common in strength and conditioning programs (1, 2, 7, 8, 11, 12, 14, 19, 20). Recent research has shown increases in strength, power, and RPD using VRT (1, 14, 20). However, optimal protocols for VRT have yet to be defined. Prior research of VRT has observed increases in strength with an array of band tensions equal to 15-35% of total load (1, 2, 17). Additionally, power increases with VRT have been observed at 20-35% of total load (1, 17), but not with 15 % of total load (2). These findings suggest $\geq 20\%$ of total load as band tension may be necessary to optimize training adaptations.

Wallace et al. (20) conducted a study in which several variables including mean force, mean power, and rate of force development were compared under free weight and 20-35 % band

1 tension conditions. It was observed that greater band tensions (35%) produced significantly
2 greater rate of force development and mean power compared to the 20% band tension and
3 free weight groups only. However, these observations occurred in an acute setting and longer
4 duration training studies need to be conducted to determine a true dose dependent
5 relationship.

6
7 Rhea et al. (14) took a different approach to applying elastic bands to a resistance training
8 protocol. Rather than matching for barbell loads, this study compared varying barbell velocity
9 conditions, with and without resistance bands. Specifically, they had one heavy / slow
10 resistance training group which was compared to controlled velocity fast VRT and traditional
11 resistance training groups. Using VRT and fast lifting speeds produced equal increases in
12 strength compared to heavy and slow resistance training, while also increasing in power more
13 than the other groups. However, despite favorable increases in strength and power, the
14 amount of band tension was never quantified. Rather, it was stated that the bands group used
15 50% 1RM as free weight and applied additional band tension.

16 Due to VRT becoming a prevalent, useful training technique, it is important to identify the
17 chronic effects of applying greater band tensions in a practical strength and conditioning
18 setting. Previous research has examined a wide range of resistances applied as band tension,
19 but to our knowledge, only one study has examined applying band tension in excess of 25%
20 1RM (20). However, this study was acute in nature. Therefore, the chronic effects of greater
21 band tensions remain to be investigated. Moreover, no research that we are aware of has
22 included VRT within the context of a traditional periodized resistance training split. Therefore,

the purpose of this study was 1) to examine the chronic effects of VRT on measures of strength and power with greater band tension than has been observed in past literature and 2) to apply elastic bands within the context of a more common daily undulating periodized resistance training protocol during the early phase of an annual plan.

Methods

Experimental Approach to the Problem

Participants were divided into two groups, either VRT or free weights only (control), based on 1RM strength ($p=0.94$ squat 1RM; $p=0.41$ bench press 1RM). All participants were required to undergo identical training protocols, but the VRT group applied band tension to the squat and bench press one out of every four workouts. Measures of body composition, strength and power were collected by a blinded researcher prior to and following the 5-week protocol during the afternoon to prevent diurnal effects on outcome measures.

Subjects

Fourteen NCAA division II male basketball players (91.8 ± 13.2 kg, 191.4 ± 12.5 cm) agreed to participate in this study. All participants were considered healthy as indicated by pre-season physicals, reporting no contraindications to high intensity resistance training, and they were

free from any injuries. All participants reported at least one year of prior experience in weight-lifting programs. Each participant agreed to refrain from any additional resistance or cardiovascular training not prescribed by the researchers. All methods and tests used in this study were reviewed and approved by the University of Tampa Institutional Review Board for research with human subjects. All participants were provided with written informed consent prior to participation.

Application of Band Tension

Elastic bands were purchased from Elite FTS (London, Ohio). It is important to note that when calculating load with the use of band tension that prior research has often calculated the amount of band tension being applied when the bands are at their greatest tension (longest length) (1, 20). Researchers then subtracted the load of the bands from the prescribed free weight. However in accordance with the natural strength curve of the athlete, band tension was applied at 30 % 1RM in addition to the prescribed weight. Our calculation of 30% was derived from the manufacturers given resistance for the bands in combination with our measured 1RM's. As such bands were at their shortest length near the zero velocity point in the exercise. When the bands were at their shortest length they provided little to no tension; with the greatest tension occurring at the top range of motion.

Strength and Body Composition Measurements

At the start of the study, subjects reported to the laboratory for baseline 1-RM testing of the full squat, bench press, and deadlift. The concentric 1RM testing began with a warm-up at a light resistance of 50% 1RM (5-10 repetitions). The load was then increased in 13.64 to 18.18 kg increments until only one successful repetition could be completed. Each participant 1-RM was determined in approximately five attempts as all 1RMs were found within these attempts. Each lift was deemed successful as described by International Powerlifting Federation rules. In the event of a failed 1-RM attempt, the weight was decreased by 4.54-9.09 kg until completion of a successful lift. Body composition was determined on a Lunar Prodigy dual x-ray absorptiometry apparatus (software version, enCORE 2008, Madison, Wisconsin, U.S.A.) prior to and following completion of the training protocol.

Power Measurements

Measurements of power taken included vertical jump height, peak power, and RPD. Vertical jump with one step was recorded from a vertec (Sports Imports, Columbus, OH). Vertical jump from a standing position, peak power and RPD were collected on a multicomponent AMTI force platform (Advanced Mechanical Technology, Inc., Watertown, MA) which interfaced with a personal computer at a sampling rate of 1000 Hz. Data acquisition software (LabVIEW, version 7.1; National Instruments Corporation, Austin, TX). Jump height on the force platform was calculated via the formula $[\frac{a \times t^2}{8}]$ where a is the acceleration because of gravity (9.81 m/s^2) and t is flight time (seconds). Peak power was calculated as the peak combination of ground

reaction force and peak velocity during the accelerated launch on the platform. Reliability of VJ height and power ranged from 0.96 to 0.97. The 3RM in the clean exercise was also used as a power measure. Vertec and force plate measurements were taken in sequence prior to the beginning of the study and 72 hours following strength assessment. The warm-up preceding vertec and force plate power measurements consisted of three sets of ten body-weight squats. Both vertec and force plate measurements utilized a best of three testing system. During all vertical jump testing, athletes were instructed to jump as high as possible.

Resistance Training Protocol

The resistance training protocol acted as the athletes' preseason training, taking place during the five weeks immediately prior to their first scrimmage. The training program consisted of a daily undulating periodized scheme composed of two power oriented sessions, one strength oriented session, and one hypertrophy oriented session per week. Athletes also participated in three sprinting sessions per week for a total of 5 weeks (table 1). Each resistance session was a full body workout consisting of compound exercises. The training program was identical between groups with the exception of the elastic bands being applied at a tension of 30% 1RM to the VRT group on one of the power sessions per week. During this time, band tension was only applied to the squat and the bench press exercises.

Statistical Analyses

All data were presented using descriptive statistics (mean $\pm$ SD). A repeated measures ANOVA was used to identify group, time, and group by time interactions for both raw data and percent change values. A Fisher LSD post-hoc was used to locate differences. Effect sizes (ES) were calculated as the difference of the baseline and post-test means divided by the average of the baseline and post-tests standard deviations. The magnitude of effect was classified by Rhea as trivial if the effect size was less than 0.25, small if the effect size was between 0.25 and 0.50, moderate if the effect size was between 0.50 and 1.0, and large if the effect size was greater than 1.0 (13). The level of significance was set at $p \leq 0.05$. Statistica software (StatSoft®, Tulsa, OK) was used to perform the statistical analyses

Results

No baseline differences were observed between groups for any measurement of strength, power, or body composition ($p \geq 0.05$). A significant group by time interaction was observed for RPD ($p=0.03$), in which RPD was greater in VRT post training than in the control group (Figure 1A). Percent increases and ESs demonstrated a much greater treatment effect for RPD in the VRT group than in the control group (Table 2).

Insert table 2

Insert figure 1

Significant time effects were observed for all other variables including squat 1RM, bench press 1RM, deadlift 1-RM, clean 3-RM, vertical jump, and lean mass (Table 2). While there were no

significant group X time interactions, the VRT group's percent changes and ESs indicate a larger treatment effect in the squat and bench press 1RM values (Figure 2A and 2C) and the vertical jump performed on the force plate and vertec. Considering the robust changes in percent change and ES, it is possible statistical power was too low to produce significant outcome measures for strength, power, and vertical jump. Effect size and percent change data were similar in all other movements.

Insert figure 2

Discussion

The purpose of this study was to examine the effects of VRT on measures of strength and RPD with greater band tension than has been observed in past literature and to apply elastic bands within the context of a more common daily undulating periodized resistance training protocol.

The primary findings of this study were that VRT with the use of elastic bands comprising 30% of the load applied to one workout per week resulted in greater changes in RPD than the control group. Moreover, VRT resulted in greater treatment effects in squat, bench press and all jumping measurements.

For athletic performance, the rate that individuals develop force and velocity has been deemed more important than their respective peak values (5, 6). Rate of power development is an expression of the ideal combination of both of these variables (5, 6). Our results demonstrated that the addition of bands to one training session per week resulted in greater RPD than a

control group (20.5% vs -12.3%), respectively. Consequently, treatment effects in both the standing and vertec jump were greater following VRT than the control group. Our results are in accordance with Rhea et al. (14) who compared the effect of heavy resistance and slow movement, lighter resistance and fast movement, or fast movements with variable resistance using bands. These researchers found greater treatment effects for vertical jump power in the variable resistance group as compared to the slow and traditional fast groups (9). Our study design was unique in that it incorporated a variable resistance day within the context of an undulating periodized training split. To our knowledge, this is the first study to examine variable resistance in this context, expanding its use in a more practical field setting.

Greater changes in RPD and jumping capacity following the use of resistance bands may be attributed to greater neuromuscular demands placed upon a given movement. Skeletal muscle is capable of producing the most amount of force near its resting length due to optimal sarcomere filaments overlap (9). With the consideration that the majority of sticking points likely exist when muscles are furthest from their resting length, it could be hypothesized that type II fibers are recruited at an accelerated rate near the bottom portion of a lift (10). The results are robust neuromuscular adaptations, which lead to greater expression of power. Following the sticking point in any exercise, a deceleration phase exists in which the muscles are not optimally contracting; rather, they are reducing force output to accommodate the need for the weight to stop (10). However as shown by Wallace et al. (20) exercises performed with band tension may be done with optimal loads for any given point through the range of motion, thus extending the range of neuromuscular adaptations throughout the entire range of motion

of the exercise. This was depicted by Andersen et al. who identified that the bar decelerates less through the whole range of motion because of the elastic resistance (1). Therefore, it can be speculated that using variable resistance can augment the rate at which power is developed during a contraction.

Strength measurements portrayed interesting results. The squat and bench press exercise both resulted in greater effect size changes than the control group (i.e. 1.42 vs 0.90 and 0.60 vs 0.19), respectively. However, no differences were observed in the deadlift between VRT and control groups (i.e. 2.04 vs 2.41, which did not receive the application of the elastic bands. This presents further evidence that elastic bands provide unique adaptations that may be specific to the task they are applied to. There are a number of possible mechanisms for the adaptations seen in our study. First, research suggests that adaptations in strength are specific to the load demands placed on an individual during a given training session. For example, Campos et al. (3) demonstrated that treatment effects for 1-RM strength increased with greater repetition maximum loading zones. While free weights alone provide robust changes in force generating capacity, they may not mechanically operate to maximize an individual's full neuromuscular potential (1). As previously described, an individual's strength is limited by a mechanical disadvantage at the beginning range of a given lift (10). For example in a squat, mechanical advantage is greatest near full extension, and least near the bottom portion of the lift (16). Thus, the majority of force and therefore acceleration required by contraction occurs near the bottom range of motion (16). As the lift continues, mechanical advantage increases and the intramuscular force requirements and acceleration of a given lift decrease. Resistance bands

1 increase tension proportional to their length. As such, the application of elastic bands can be
2 used to accommodate an athlete's strength curve (14). In support, research from Wallace and
3 colleagues (20) have demonstrated that the application of resistance bands increases peak and
4 average force, while extending the range of acceleration of a given lift.

5
6 Our research agreed with Andersen (1) who found that NCAA athletes, similar to our own,
7 demonstrated greater effect size increases in squat strength with resistance bands ($ES = 0.47$)
8 as compared to the free weight only group ($ES = 0.2$) following 7 weeks of training. Our results
9 also agreed with Rhea (14) who also found greater treatment effects in the squat in NCAA
10 athletes following 12 weeks of resistance training when using resistance bands ($ES = 1.10$) as
11 compared to a free weight heavy /slow (1.08) and light / fast ($ES=0.38$) group. However, while
12 our study was only 5 weeks in duration our effect sizes on changes in the squat are the largest
13 to date in a resistance band study ($ES = 1.42$ in VRT compared to 0.92 in resistance training
14 only). These results may suggest that the application of variable resistance as part of a daily
15 undulated periodized training split is ideal for neuromuscular adaptations. It may also be
16 attributed to the increased amount of band tension (30%) compared to previous literature, and
17 future research should further examine even greater band tension. A third explanation for our
18 results may lie within the method of band application. Specifically, Rhea et al. (14) matched
19 velocity between fast resistance and fast variable resistance groups. However, Andersen et al.
20 (1) matched groups for average workload. Specifically, when an athlete was at the bottom and
21 top portion of a lift they had 10 % lower and 10 % higher loading than the free weight only
22 group, respectively. Our study was novel in the fact that we added band tension to the normal

load used on a power day. Our rationale was to fully accommodate the strength curve, by maintaining the force requirements of the bottom portion of a lift, while simultaneously increasing the requirements of the top range of the lift. Future research will need to delineate which method is ideal chronically for both strength and power gains. It might be postulated that greater eccentric loading in our protocol would be ideal for strength. Specifically, research suggests a preferential recruitment of type II fibers with eccentric as compared to concentric loading (4). As such, Andersen (1) postulates that the relatively overloaded eccentric portion of a lift with added resistance bands may challenge the neuromuscular system to recruit a larger population of type II fibers, ultimately leading to greater strength adaptations. It is possible that these athletes suffered a detraining effect prior to beginning this study, as their training began following their summer break, during which they were unmonitored. Our research was limited in its duration of five weeks, and it was also limited as we did not have the capabilities to measure upper body power. Future research should take these variables into account.

Practical applications

Our results suggest that using VRT within the context of a daily undulating periodized protocol during the early phase of an annual plan can increase RPD, vertical jump height, and strength. When applied to select workouts, variable resistance adds flexibility to a training program. Specifically, an athlete can train with variable resistance one day out of the week and then train at higher intensities for strength or skeletal muscle hypertrophy on other days of the week.

1 Coaches can also implement elastic bands at a tension of 30% 1RM to improve performance
2 related variables, such as strength. Finally, for strength and power oriented athletes or
3 practitioners it is conceivable that VRT can be used to overcome a stagnation of progress and
4 break plateaus.

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3 This study was not funded by any company or grant.

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- 1
- 2 Conflict of Interest Statement
- 3
- 4 The authors declare that they have no conflict of interest

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	Monday	Tuesday	Wednesday	Thursday	Friday
AM	Squat (w/bands in VRT) Bench Press (w/bands in VRT) Deadlift Shoulder Press Weeks 1 & 4: 3x5 40% 1RM Weeks 2 & 5: 3x4 50% 1RM Week 3: 3x3 60% 1RM	Squat Bench Press Pull-up Deadlift Shoulder Press Bent Row Weeks 1 & 4: 3x5 85% 1RM Weeks 2 & 5: 3x3 90% 1RM Week 3: 3x2 95% 1RM	Long distance sprints	Clean Half Squat Depth Jumps Plyo Pushups Jumping Lunges Weeks 1 & 4: 3x5 clean @ 80% 1RM half squat @ 40% 1RM Weeks 2 & 5: 3x4 clean @ 85% 1RM half squat @ 50% 1RM Week 3: 3x3 clean @ 90% 1RM half squat @ 60% 1RM	medium distance sprints
PM	Short distance sprints				Squat Bench Press Lunge Pullups Shoulder Press Bent over Row Week 1 & 4: 3x12 65% 1RM, 1m rest Week 2 & 5: 3x10 70% 1RM, 90s rest Week 3: 3x8 75% 1RM, 2m rest

Table 1. Exercise protocol was the same per condition. The VRT group received application of band tension on the squat and bench press on Mondays. Sets and repetitions are reported as (number of sets)x(number of repetitions).

Variable	Week 0 Means	Week 5 Means	Pooled Standard Deviation	Effect Size	%Change
VRT Squat Strength(kg)*	127.60	159.09	54.42	1.42	26.63
Squat Strength(kg)	129.17	149.24	54.63	0.91	20.03
VRT Bench Press Strength(kg)*	97.40	104.87	24.57	0.60	7.72
Bench Press Strength(kg)	109.85	112.88	34.45	0.19	3.30
VRT Deadlift Strength(kg)	151.62	192.53	50.80	2.04	28.24
Deadlift Strength(kg)	150.76	196.59	46.44	2.41	30.82
VRT Clean Strength(kg)	82.14	90.59	28.78	0.74	11.34
Clean Strength(kg)	76.14	89.77	34.02	1.07	19.01
VRT Vertec VJ Height(in)	35.67	37.50	3.42	0.56	6.33
Vertec VJ Height(in)	31.33	32.42	2.25	0.46	4.13
VRT Force Plate VJ Height(in)	24.69	26.37	4.61	0.38	7.52
Force Plate VJ Height(in)	22.82	22.60	4.17	-0.04	-1.79
VRT Maximum Power(W)	7033.20	7459.67	640.03	0.70	6.57
Maximum Power(W)	7607.77	7184.58	1280.47	-0.30	-5.35
VRT RPD(W/s)	24212.71	29546.29	2693.74	0.99	20.54
RPD(W/s)	25622.50	21994.50	6683.74	-0.57	-12.39
VRT Lean Mass(kg)	72.34	74.79	2.64	0.92	3.40
Lean Mass(kg)	75.38	76.75	4.56	0.27	1.88

Table 2. * denotes exercise which received treatment of elastic bands.

Figure 1

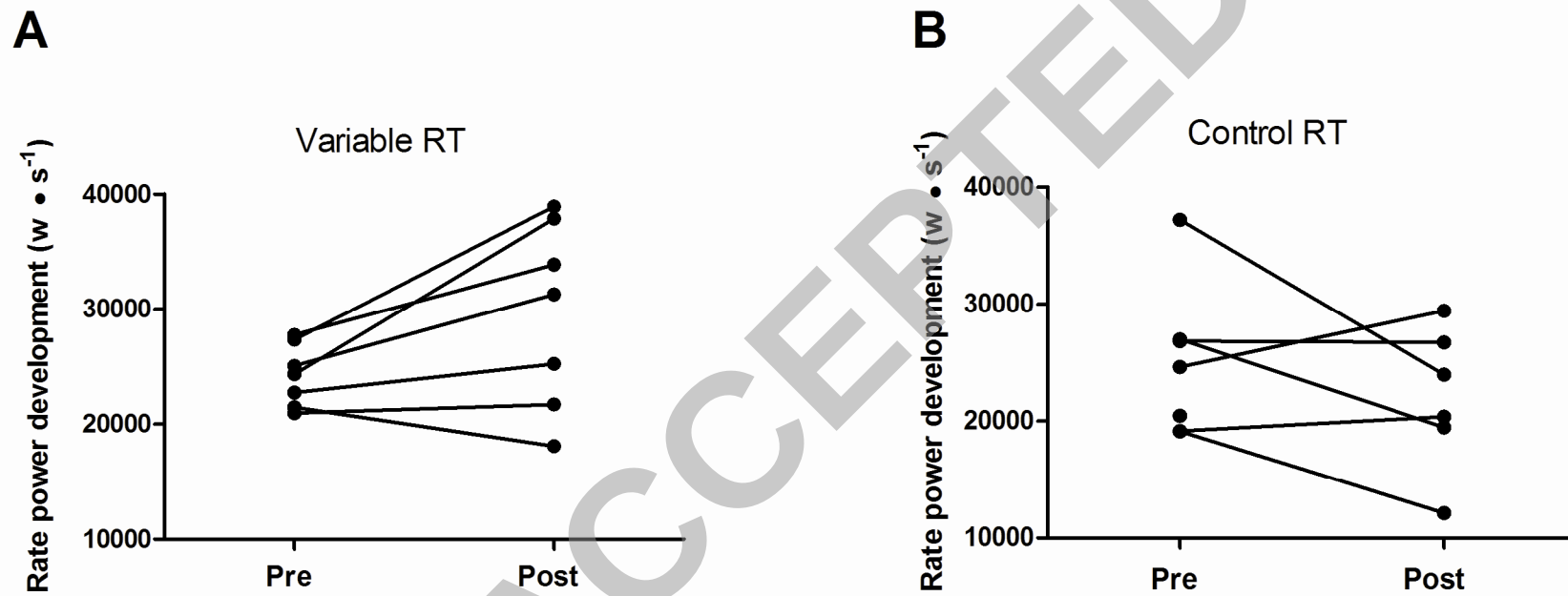


Figure 2

